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Modelling of diffuse solar fraction with multiple predictors

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ABSTRACT

For some locations both global and diffuse solar radiation are measured. However, for many locations, only global radiation is measured, or inferred from satellite data. For modelling solar energy applications, the amount of radiation on a tilted surface is needed. Since only the direct component on a tilted surface can be calculated from direct on some other plane using trigonometry, we need to have diffuse radiation on the horizontal plane available. There are regression relationships for estimating the diffuse on a tilted surface from diffuse on the horizontal. Models for estimating the diffuse on the horizontal from horizontal global that have been developed in Europe or North America have proved to be inadequate for Australia [13]. Boland et al. [2] developed a validated model for Australian conditions. Boland et al. [3] detailed our recent advances in developing the theoretical framework for the use of the logistic function instead of piecewise linear or simple nonlinear functions and was the first step in identifying the means for developing a generic model for estimating diffuse from global and other predictors. We have developed a multiple predictor model, which is much simpler than previous models, and uses hourly clearness index, daily clearness index, solar altitude, apparent solar time and a measure of persistence of global radiation level as predictors. This model performs marginally better than currently used models for locations in the Northern Hemisphere and substantially better for Southern Hemisphere locations. We suggest it can be used as a universal model.

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1. Introduction

There are two objectives for performing the analysis in this paper. Models for estimating the diffuse on the horizontal from horizontal global that have been developed in Europe or North America have proved to be inadequate for Australia [13]. This may be partially explained by observing the significantly higher proportion of values with high clearness index and corresponding low values of diffuse fraction in Adelaide Australia, than Bracknell, UK as seen in Fig. 1. Balling and Idso [1] observed higher levels of stratospheric sulphate aerosol effects in the Northern Hemisphere than in the Southern Hemisphere and Gueymard [6] states *"Atmospheric turbidity characterizes the extinction effect of aerosols on solar radiation. It is well known that turbidity is the most important extinction affecting the direct and diffuse component of solar radiation under clear skies and that, as turbidity increases, the former decreases and the latter increases in about the same proportions"*. The phenomenon of higher turbidity in the Northern Hemisphere may explain the lower proportion of high clearness values observed in those locations.

Boland et al. [2] developed a validated model for Australian conditions, using a logistic function instead of piecewise linear or simple nonlinear functions. Recently, Jacovides et al. [7] have verified that this model performs well for locations in Cyprus. The analysis by Boland et al. [3] included using a "moving average" technique to demonstrate the form of the relationship, which corresponds well to a logistic relationship. This research further detailed our recent advances in developing the theoretical framework justifying the use of the logistic function instead of piecewise linear or simple nonlinear functions and the first step in identifying the means for developing a generic model for estimating diffuse from global and other predictors. The single predictor generic model for diffuse fraction

$$d = \frac{I_{\text{diffuse}}}{I_{\text{global}}} \tag{1}$$

takes the hourly clearness index

$$k_t = \frac{I_{\text{global}}}{H_0} \tag{2}$$

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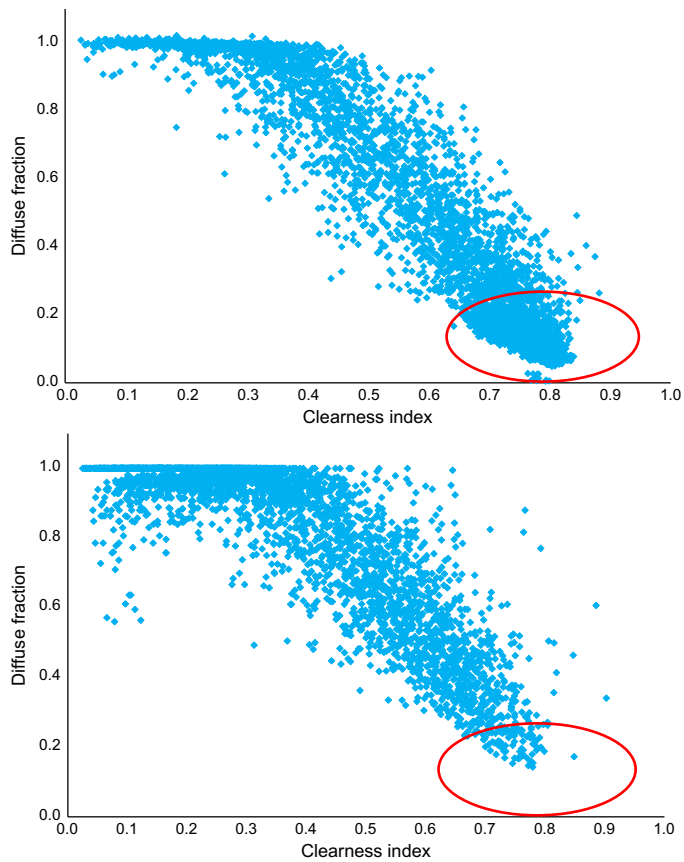

Fig. 1. Raw data for Adelaide and Bracknell.

as the sole predictor. where I_{global}, I_{diffuse} and H_0 are the global, diffuse and extraterrestrial radiation integrated over the hour in question. The generic equation with one predictor is given by [3].

$$d = \frac{1}{1 + e^{-5.0033+8.6025k_t}} \tag{3}$$

We have made significant advances in both the intuitive and theoretical justification of the use of the logistic function. Fig. 2 illustrates the need for an extension of this methodology to include multiple predictors so that we better describe the spread of the data. This is a first step in identifying the means for developing a logistic model for estimating diffuse from global and then

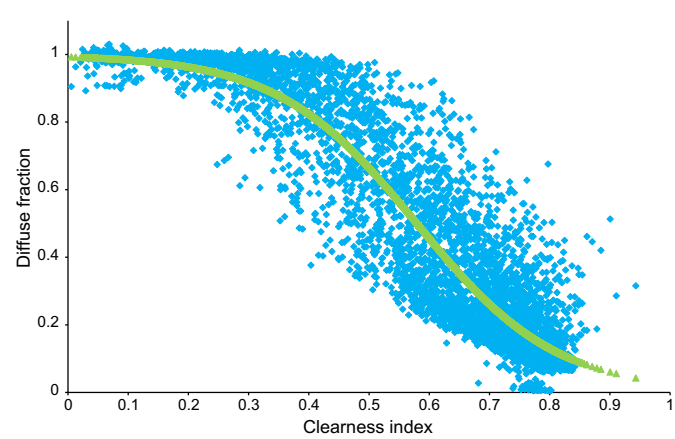

Fig. 2. Adelaide data with the single predictor generic model overlaid.

constructing a generic model, since the main problem we are trying to deal with is the lack of transportability of models constructed for other climates. Examples for both Northern and Southern Hemisphere locations will be presented. We will then move on to a generic multiple predictor model.

2. Multiple predictor generic model

Reindl et al. [11] presented a comprehensive study of the prediction of the diffuse fraction of solar radiation from other ground variables, including clearness index, relative humidity, solar altitude angle and so on (a total of 28). The clearness index is the proportion of extraterrestrial irradiation reaching a location and thus is a measure of 'cloudiness'. However, we will consider only one of their models. They found that most of the possible predictors gave insignificant benefit to the prediction.

The four that they used in the final model included relative humidity and ambient temperature which are measured variables. Our objective is to be able to predict the diffuse fraction with as few measured predictors as possible. The model could then be used to predict the diffuse (and thus direct) radiation with only having global measured or inferred from satellite data. Thus we will base our comparisons between our work and the model of Reindl et al. [11], which uses only clearness index and solar altitude (solar altitude being a calculated rather than measured variable) and the models developed by Skartveit et al. [12] and Perez et al. [10]. In fact, if one examines the results from Reindl's work, an analysis of the efficacy of adding the extra two variables may well argue that they do not add sufficiently to the predictability to consider them. The form of Reindl's model we will be using for comparison is

$$\begin{aligned} d &= \eta_1 + \gamma_1 k_t + \delta_1 \sin\alpha \quad 0 \le k_t \le 0.3 \quad d \le 1.0 \\ d &= \eta_2 + \gamma_2 k_t + \delta_2 \sin\alpha \quad 0.3 < k_t < 0.78 \quad 0.1 \le d \le 0.97 \\ d &= \eta_3 + \gamma_3 k_t + \delta_3 \sin\alpha \quad k_t \ge 0.78 \quad 0.1 \le d \end{aligned} \tag{4}$$

Here α is the solar altitude and η_i, γ_i and δ_i are the parameters to be determined.

The Skartveit model is very complicated, but like Reindl's it is constrained over different sub-intervals of the clearness index. It uses three explanatory variables; clearness index, solar altitude and a predictor which is called the hourly variability index (σ_3) and is defined as the "root mean squared deviation between the 'clear sky' index of the hour in question (ρ) and, respectively, the preceding hour (ρ_{t-1}) and the succeeding hour (ρ_{t+1})". This model thus also has the clearness index as its sole measured predictor.

Another model used for comparison is that of Perez et al. [10] which estimates direct solar irradiance from which the diffuse radiation can be calculated. Perez presents a 4-dimensional model which utilises the following variables:

- Z – solar zenith angle
- Kt' – a zenith angle-dependent expression of the clearness index
- $\Delta Kt'$ – stability index
- W – atmospheric precipitable water

but can be reduced to 3D mode in the absence of W. Perez's model modifies the DISC model of Maxwell [9] by an insolation-dependent factor as shown in Equation 5.

$$I = I_{\text{DISC}} \times X(Kt', Z, W, \Delta Kt') \tag{5}$$

where I_{DISC} is the direct normal irradiance estimated by the DISC model and $X(Kt', Z, W, \Delta Kt')$ is a coefficient function of the four insolation parameters. This coefficient is obtained from a

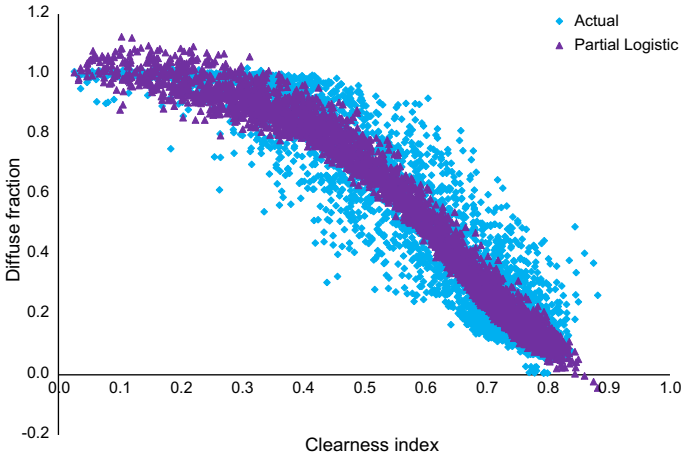

Fig. 3. Adelaide data with the partial logistic multiple predictor model overlaid.

4-dimensional look up table consisting of a $6 \times 6 \times 5 \times 7$ matrix based on bins for each of the four variables.

As mentioned previously, Reindl et al. [11] identified 28 possible predictor variables and through statistical analysis determined that four of these (clearness index, ambient temperature, the sine of the solar angle and relative humidity) gave the best results. We will consider the solar angle α (in degrees) out of this grouping, as well as a number of other possible predictors. We also consider apparent solar time (AST) as well as solar angle since it, unlike the altitude, is asymmetric about solar noon, and this may aid in explaining differences in the atmosphere between morning and afternoon. Skartveit et al. [12] and Gonzales and Calbó [5] have also included a predictor to account for variability. We consider a similar type of variable, but in a different form. Instead of using a measure of how much the present hour's clearness differs from surrounding hours, we take a point from Erbs et al. [4], where, after determining the dependence of the hourly diffuse fraction on clearness index, they take the error, or residual values, and model them as a first order autoregressive model. This serial dependence concept has intuitive appeal, since it could be argued that there is some inertia in the atmosphere that can be picked up in this manner. It could be that this inertia can be encapsulated in using values of the lagged clearness index as a predictor. However, since we are not attempting to forecast the diffuse fraction, we take "persistence" as an extra predictor which is an average of both a lag and a lead of the clearness index:

$$\psi = \begin{cases} \frac{k_{t-1}+k_{t+1}}{2} & \text{sunrise} < t < \text{sunset} \\ k_{t+1} & t = \text{sunrise} \\ k_{t-1} & t = \text{sunset} \end{cases} \tag{6}$$

As well, we consider that there may well be a case for the daily clearness index (K_t) to be used as a predictor – the whole day may have a common characteristic. Daily clearness index is calculated as

$$K_t = \frac{\sum_{j=1}^{24} I_{\text{global}_j}}{\sum_{j=1}^{24} H_{0_j}} \tag{7}$$

where I_{global_j} is global radiation at hour j and H_{0_j} is extraterrestrial radiation at hour j.

Note that what we are trying to do is to find as many possible predictors as we can of a type that requires as little as possible recorded data. The number of sites in Australia that are recording even global solar radiation at sub-diurnal time scales is dropping, with a greater dependence on satellite inferred daily totals. A daily profile can be inferred from that data, but then diffuse values will have to be estimated from a model relying possibly on very few measured values.

An extension of the single predictor model with the additional predictors in linear format

$$d = \frac{1}{1 + e^{\beta_0 + \beta_1 k_t}} + \beta_2 \text{AST} + \beta_3 \alpha + \beta_4 K_t + \beta_5 \psi \tag{8}$$

did not constrain the endpoints of the model (see Fig. 3), resulting in infeasible values, and so the model that we are exploring takes the full logistic form as shown in Eq. (9) which naturally bounds the endpoints without need for further constraints.

$$d = \frac{1}{1 + e^{\beta_0 + \beta_1 k_t + \beta_2 AST + \beta_3 \alpha + \beta_4 K_t + \beta_5 \psi}} \tag{9}$$

To build the multiple predictor model, all parameters need to be estimated including β_0 and β_1. We have accessed data from seven locations worldwide. The information provided for the data is as follows:

- Adelaide and Darwin data is hourly integrated data which was provided by the Australian Bureau of Meteorology.
- Bracknell, Lisbon, Macau, Maputo and Uccle data sets were collected by different institutions at different times which were supplied as data files with no metadata associated.

We performed regression analysis for the significance of the parameters and have found that, with the exception of those marked with an asterisk in Table 1, all were found to be highly significant at the 0.05 level. Although some parameters were not significant at all locations, we decided to include them in the model

Table 2
Summary of Partial F-test.

Variables	SS_R	$\frac{SS_E}{n-2}$	F	p
No Kt or ψ	2607			
Add ψ	2626	0.0099	1959	0.000
Add K_t	2639	0.0096	3272	0.000

Table 1
Multiple predictor logistic model parameters for all locations.

Variable	Southern hemisphere			Northern hemisphere			
	Adelaide	Darwin	Maputo	Bracknell	Lisbon	Macau	Uccle
Constant	−6.0007	−4.6976	−6.4431	−5.0108	−5.0812	−5.1661	−4.8720
k_t	8.0671	5.4582	7.1617	6.1307	6.1169	6.3027	7.0306
AST	0.0024*	0.0007	0.0375	−0.0135*	0.0027*	−0.0037	0.0035
α	−0.0135	−0.0013	−0.0088	−0.0068	−0.0093	−0.0088	−0.0215
K_t	1.2590	1.8179	1.9497	1.3986*	1.3972	2.1174	1.8415
ψ	1.5984	1.3377	1.7943	1.3606	1.5143	1.4891	1.1425

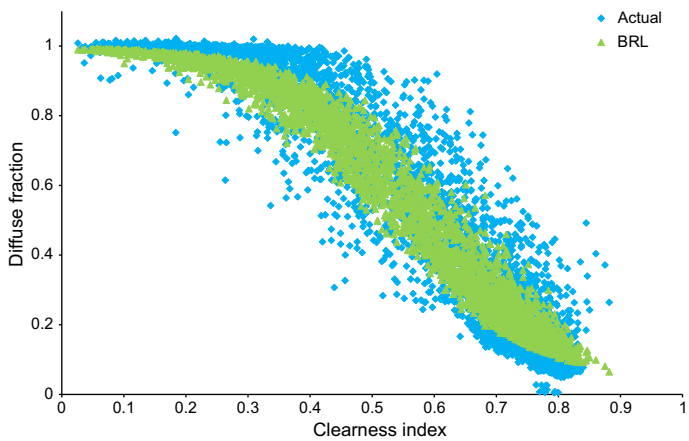

Fig. 4. Adelaide data with the BRL model overlaid.

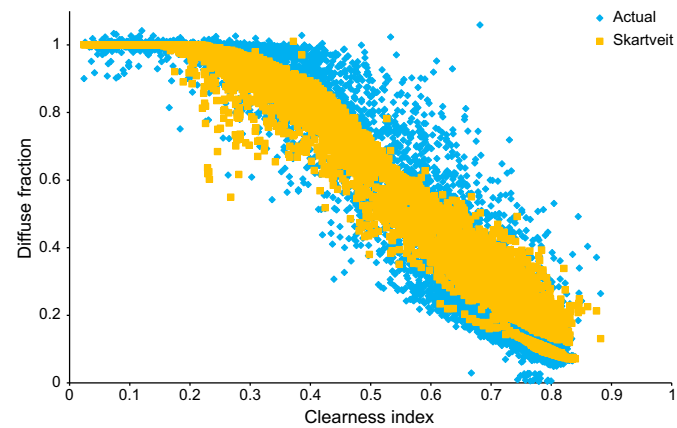

Fig. 6. Adelaide data with the Skartveit model overlaid.

as initial inspection indicated that even a small contribution improved the spread of the data and there is no added difficulty in estimating these values as we utilised the minimum least squares method in Solver (an optimisation tool in Excel®) which quickly estimates the optimal values for all parameters.

The parameter estimates given in Table 1 are not so dissimilar as to make us believe that we have to have a separate model for each separate location. We developed the multiple predictor generic logistic model by a robust method for estimating the parameters which involved amalgamating the data from all seven locations and then minimising the residual sum of squares using Solver for the complete data set. The generic multiple predictor logistic model, hereafter to be known as the Boland–Ridley–Lauret (BRL) model, is

$$d = \frac{1}{1 + e^{-5.38+6.63k_t+0.006\mathrm{AST}-0.007\alpha+1.75K_t+1.31\psi}} \tag{10}$$

To further support the inclusion of all predictors in the final BRL model we have used the partial F-test. The partial F-test determines whether the addition of the $k+1$ variable improves the model, given the inclusion of the previous k variables. The calculation of the statistic is

$$F_{\alpha,r,n-k} = \frac{SS_R(k+1) - SS_R(k)}{MS_E} \tag{11}$$

$$\text{Where } MS_E = \frac{SS_E(k+1)}{n-2} \tag{12}$$

$$SS_E = \sum_{i=1}^{n}\left(d - \widehat{d}\right)^2 \tag{13}$$

$$SS_R = \sum_{i=1}^{n}\left(\widehat{d} - \overline{d}\right)^2 \tag{14}$$

Here $\widehat{d}$ is the model estimate for diffuse fraction d, $\overline{d}$ is the mean of the actual diffuse fraction, r is the number of additional regressors, n is the number of observations and k is the number of regressors already in the model. We test the hypothesis that

$$H_0 : \beta_{k+1} = 0$$
$$H_1 : \beta_{k+1} \neq 0$$

We have set $r = 1$ in order to corroborate separately the inclusion of K_t and ψ in the final model. We can see in Table 2, which summarises the calculations for the Partial F-test, that with a p-value of 0.000 both K_t and ψ are highly significant and so we reject the null hypothesis and conclude that these two variables should be included in the model. Such a low p-value is not unexpected because, although the difference in the SS_R values does not appear to be significant, it is the fact that with such a large data set ($n = 25{,}299$) the MS_E is very small and so F is large.

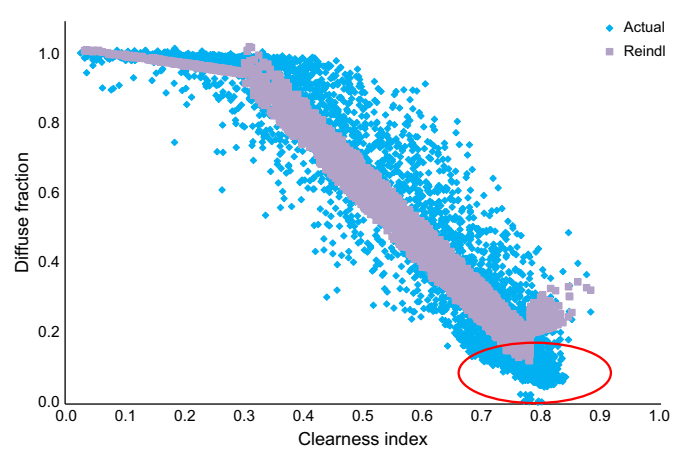

Fig. 5. Adelaide data with the re-estimated Reindl model overlaid.

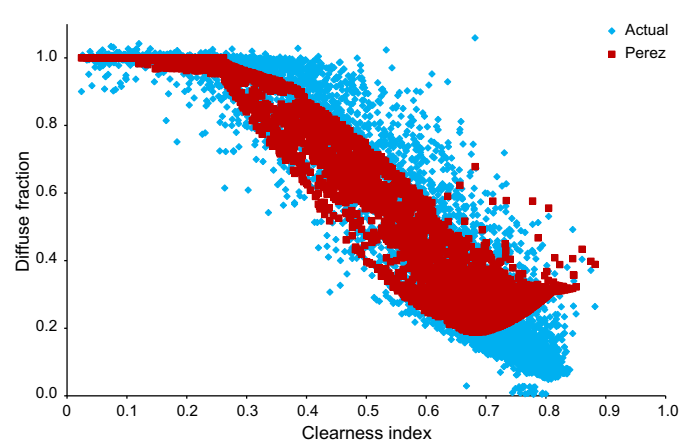

Fig. 7. Adelaide data with the Perez model overlaid.

Table 3
Error analysis for the BRL, Reindl, Skartveit and Perez models – all against the actual data for each location.

		Southern Hemisphere			Northern Hemisphere			
		Adelaide	Darwin	Maputo	Bracknell	Lisbon	Macau	Uccle
MeAPE	BRL	17.15	21.13	20.37	5.52	13.87	7.11	6.37
	Reindl	20.22	31.80	29.04	5.62	16.14	7.90	7.31
	Skartveit	18.91	25.62	21.35	5.82	17.08	6.91	6.69
	Perez	31.73	41.78	24.38	6.82	23.48	7.75	7.14
MeBE	BRL	−0.023	−0.144	−0.084	0.023	0.071	0.007	0.008
	Reindl	−0.033	−0.295	−0.213	0.023	0.003	−0.007	−0.003
	Skartveit	−0.033	−0.206	−0.122	0.007	0.008	0.000	0.000
	Perez	−0.103	−0.394	−0.102	0.014	−0.019	0.000	0.000

Table 4
BIC($\times 10^4$) for the BRL, Reindl and Skartveit models – all against the actual data for each location.

	Southern Hemisphere			Northern Hemisphere				Number of parameters
	Adelaide	Darwin	Maputo	Bracknell	Lisbon	Macau	Uccle	
BRL	−1.77	−1.15	−1.57	−1.58	−1.60	−1.52	−1.64	6
Reindl	−1.71	−1.06	−1.44	−1.48	−1.57	−1.47	−1.62	9
Skartveit	−1.74	−1.07	−1.54	−1.51	−1.62	−1.48	−1.67	>20

3. Validation

A graphical comparison for the Adelaide data of the fit of the BRL model overlaid against the actual data is shown in Fig. 4 where we can observe a very good fit to the entire data set. Fig. 5 shows the fit of Reindl's model where we see that, in the area circled, Reindl's model does not extend far enough into the "high clearness–low diffuse" values. Skartveit's model, as shown in Fig. 6, seems to fit the data reasonably well. Perez's model (Fig. 7) seems to fit the data reasonably well but again does not extend far enough into the high clearness index region.

After inspecting results for all the other locations studied, it appears that Reindl's and Skartveit's models do not perform as well for locations in the Southern Hemisphere, whereas the BRL model seems to fit the data more fully and evenly for all locations.

We now move onto formal error analysis. We will examine the median of the absolute percentage error (MeAPE) as a measure of the size of the respective errors and the median of the bias error (MeBE) to determine whether any particular model is more biased than another. The median rather than mean was used in the error analysis because we found that the residuals for all models were not normally distributed.

From Table 3, which summarises the MeAPE and MeBE for the BRL, Reindl, Skartveit and Perez models against the actual data, we first examine the Southern Hemisphere locations. When comparing the errors for these locations we see that the BRL model is performing better than all the other models for both the absolute error and the bias error. Next we inspect the Northern Hemisphere locations. Here we found that the BRL model generally performed better than the other models in terms of the absolute error and was equitable to them in terms of the bias error.

Table 5
BIC($\times 10^4$) for the BRL and Perez models – all against the actual data for each location.

	Southern hemisphere			Northern hemisphere			
	Adelaide	Darwin	Maputo	Bracknell	Lisbon	Macau	Uccle
BRL	−2.57	−0.83	−1.09	−1.41	−1.23	−0.43	−1.57
Perez	−2.33	−0.56	−0.90	−1.27	−0.94	−0.40	−1.46

As a further validation of the BRL model we calculated the Bayesian Information Criterion (BIC) to demonstrate the advantage of our more parsimonious model. BIC not only measures the difference between real and predicted values, but also adds a penalty for each extra parameter requiring estimation.

$$\mathrm{BIC} = n\ln\left(\frac{\mathrm{RSS}}{n}\right) + k\ln(n) \tag{15}$$

where RSS is the residual sum of squares, n is the number of data points and k is the number of parameters to be estimated. We have shown the BIC for the BRL, Reindl and Skartveit models but because it is difficult to identify how many parameters are to be estimated with Skartveit's model we have estimated k to be at least 20. We have performed separate BIC analysis for comparing the BRL and Perez models because with each of Reindl, Skartveit and Perez there are a different range of values of k_t for which each of those models are not valid. Further, we have assumed that the number of parameters for Perez's model is 6, that is, 4 values for Kt', Z, W, $\Delta Kt'$ and 2 values for I_{global} and solar zenith angle from the *DISC* model.

When comparing models a lower value of BIC indicates a better model. We can see in Table 4 that the BRL model is again

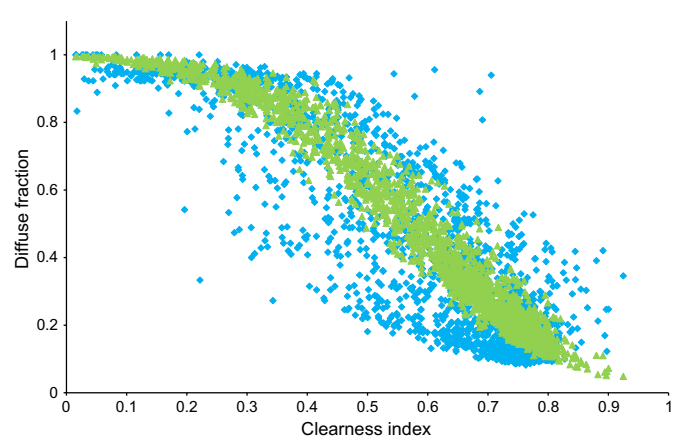

Fig. 8. Gillot data with the BRL model overlaid.

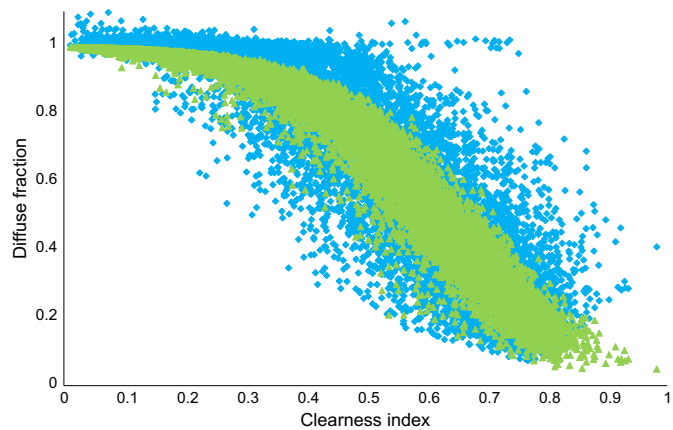

Fig. 9. Camborne data with the BRL model overlaid.

performing better than either Reindl or Skartveit for the Southern Hemisphere locations and is as good or better for the Northern Hemisphere sites. Additionally, when we inspect the BIC values in Table 5, we can see that the BRL model outperforms Perez's model for all locations in both hemispheres.

3.1. Out of sample validation

We now look at how well the model performs at sites not used in formulating the model. The two locations used are Gillot and Camborne. Gillot is a low latitude site on Rèunion Island in the Southern Hemisphere. Camborne is in England, a Northern Hemisphere site. Figs. 8 and 9 illustrate that the BRL model fits the data well for both locations. Gillot has a MeAPE of 23.97% which is of the same order as the average MeAPE for the Southern Hemisphere locations of 19.55%. Likewise, the Northern Hemisphere average MeAPE is 8.22% and our test location, Camborne, is equitable with 7.90%. This demonstrates that the BRL model performs equally well in both hemispheres. The MeBE's for Gillot and Camborne are −0.08 and 0.03, respectively, which are both on a par with the average bias errors, −0.09 and 0.03, presented for the Northern and Southern Hemisphere locations used to form the model.

4. Conclusions

We have demonstrated a method for constructing a multiple variable logistic model for diffuse solar fraction. The direct radiation can then be estimated using the formula $I_{DN} = \frac{I_{\text{global}} - I_{\text{diffuse}}}{\sin\alpha}$. When examining the absolute and bias errors we show that the BRL model performs equally well for locations from both hemispheres. However, when examining the Bayesian Information Criterion we have shown that our more parsimonious model performs consistently better in both hemispheres.

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